Appendix: Differentially Private Mechanisms using Enhanced Direction-Oriented Smooth Sensitivity

In this document, we provided the proofs of theorems and lemmas in the main paper.

Theorem 1. *For all $i \in [m]$, $\tilde{S}^{*\,i+}_{f,\beta}$ and $\tilde{S}^{*\,i-}_{f,\beta}$ are enhanced direction-oriented β-smooth upper bound on LS^{i+}_f and LS^{i-}_f. In addition, for all $x \in D^n$, $\tilde{S}^{*\,i\pm}_{f,\beta}(x) \le \tilde{S}^{\,i\pm}(x)$ holds for every enhanced direction-oriented β-smooth upper bound $\tilde{S}^{\,i+}$ and $\tilde{S}^{\,i-}$ on LS^{i+}_f and LS^{i-}_f.*

Proof. From the definition, $\forall x \in D^n : \tilde{S}^{*\,i\pm}_{f,\beta}(x) \ge LS^{i\pm}_f(x)$ holds. We then show that $\forall x, y \in D^n$, $d(x,y) = 1$: $\tilde{S}^{*\,i\pm}_{f,\beta}(x) \le e^{\beta} \cdot \tilde{S}^{*\,i\pm}_{f,\beta}(y) \ \wedge\ \tilde{S}^{*\,i\pm}_{f,\beta}(x) \le e^{\beta} \cdot \tilde{S}^{*\,i\mp}_{f,\beta}(y)$.

(i) When $\tilde{S}^{*\,i+}_{f,\beta}(x) = LS^{i+}_f(x)$:

$$\begin{aligned}\tilde{S}^{*\,i\pm}_{f,\beta}(y) \ge LS^{i+}_f(x) \cdot e^{-\beta \cdot d(x,y)} &= e^{-\beta} \cdot LS^{i+}_f(x) \\ &= e^{-\beta} \cdot \tilde{S}^{*\,i+}_{f,\beta}(x).\end{aligned}$$

(ii) When $\tilde{S}^{*\,i+}_{f,\beta}(x) = LS^{i-}_f(x) \cdot e^{-2\beta}$:

$$\tilde{S}^{*\,i\pm}_{f,\beta}(y) \ge LS^{i-}_f(x) \cdot e^{-\beta} > \tilde{S}^{*\,i+}_{f,\beta}(x) > e^{-\beta} \cdot \tilde{S}^{*\,i+}_{f,\beta}(x).$$

(iii) When $\tilde{S}^{*\,i+}_{f,\beta}(x) = \max\left(LS^{i+}_f(x'), LS^{i-}_f(x')\right) \cdot e^{-\beta \cdot d(x',x)}$ for some $x' (\ne y) \in D^n$:

$$\begin{aligned}\tilde{S}^{*\,i\pm}_{f,\beta}(y) &\ge \max\left(LS^{i+}_f(x'), LS^{i-}_f(x')\right) \cdot e^{-\beta \cdot d(x',y)} \\ &\ge \max\left(LS^{i+}_f(x'), LS^{i-}_f(x')\right) \cdot e^{-\beta \cdot (d(x',x)+1)} \\ &\quad [\because d(x',y) \le d(x',x) + d(x,y)] \\ &= e^{\beta} \cdot \max\left(LS^{i+}_f(x'), LS^{i-}_f(x')\right) \cdot e^{-\beta \cdot d(x',x)} \\ &= e^{-\beta} \cdot \tilde{S}^{*\,i+}_{f,\beta}(x).\end{aligned}$$

(iv) When $\tilde{S}^{*\,i+}_{f,\beta}(x) = \max\left(LS^{i+}_f(y), LS^{i-}_f(y)\right) \cdot e^{-\beta \cdot d(y,x)}$:

(a) When $LS^{i+}_f(y) \ge LS^{i-}_f(y)$:

$$\begin{aligned}\tilde{S}^{*\,i+}_{f,\beta}(y) \ge LS^{i+}_f(y) > LS^{i+}_f(y) \cdot e^{-\beta} \\ = \tilde{S}^{*\,i+}_{f,\beta}(x) > e^{-\beta} \cdot \tilde{S}^{*\,i+}_{f,\beta}(x),\end{aligned}$$

$$\tilde{S}^{*\,i-}_{f,\beta}(y) \ge LS^{i+}_f(y) \cdot e^{-2\beta} = e^{-\beta} \cdot \tilde{S}^{*\,i+}_{f,\beta}(x).$$

(b) When $LS^{i+}_f(y) < LS^{i-}_f(y)$:

As in case (a), $\tilde{S}^{*\,i\pm}_{f,\beta}(y) \ge e^{-\beta} \cdot \tilde{S}^{*\,i+}_{f,\beta}(x)$.

From (i) to (iv), $\forall x, y \in D^n$, $d(x,y) = 1 : \tilde{S}^{*\,i+}_{f,\beta}(x) \le e^{\beta} \cdot \tilde{S}^{*\,i\pm}_{f,\beta}(y)$ holds. In a similar manner, we can show that $\forall x, y \in D^n$, $d(x,y) = 1 : \tilde{S}^{*\,i-}_{f,\beta}(x) \le e^{\beta} \cdot \tilde{S}^{*\,i\pm}_{f,\beta}(y)$ holds.

Thereafter, we show that for all $x \in D^n$, $\tilde{S}^{*\,i\pm}_{f,\beta}(x) \le \tilde{S}^{\,i\pm}(x)$. From the definition,

$$\tilde{S}^{\,i+}(x) \ge LS^{i+}_f(x),$$

$$\begin{aligned}&\forall y \in D^n,\ d(x,y) = 1 : \\ &\tilde{S}^{\,i+}(x) \ge e^{-\beta} \cdot \tilde{S}^{\,i\pm}(y) \ge e^{-2\beta} \cdot \tilde{S}^{\,i-}(x) \ge LS^{i-}_f(x) \cdot e^{-2\beta} \\ &\wedge\ \tilde{S}^{\,i+}(x) \ge e^{-\beta} \cdot \tilde{S}^{\,i\pm}(y) \ge e^{-\beta} \cdot LS^{i\pm}_f(y).\end{aligned}$$

Here, we assume that $\forall x', y \in D^n$, $d(x',y) = k : \tilde{S}^{\,i+}(x') \ge \max\left(LS^{i+}_f(y), LS^{i-}_f(y)\right) \cdot e^{-\beta \cdot d(y,x')}$ for some $k (> 0)$. Because for any $x, y \in D^n$ s.t. $d(x,y) = k+1$, $\exists x'$ satisfying $d(x,x') = 1 \ \wedge\ d(x',y) = k$,

$$\begin{aligned}\tilde{S}^{\,i+}(x) &\ge e^{-\beta} \cdot S^{\,i+}(x') \\ &\ge e^{-\beta} \cdot \max\left(LS^{i+}_f(y), LS^{i-}_f(y)\right) \cdot e^{-\beta \cdot d(y,x')} \\ &= \max\left(LS^{i+}_f(y), LS^{i-}_f(y)\right) \cdot e^{-\beta \cdot (d(y,x')+1)} \\ &= \max\left(LS^{i+}_f(y), LS^{i-}_f(y)\right) \cdot e^{-\beta \cdot d(y,x)}.\end{aligned}$$

Consequently,

$$\begin{aligned}&\forall x, y \in D^n, x \ne y :\ \tilde{S}^{\,i+}(x) \ge LS^{i+}_f(x) \\ &\quad \wedge\ \ \tilde{S}^{\,i+}(x) \ge LS^{i-}_f(x) \cdot e^{-2\beta} \\ &\quad \wedge\ \ \tilde{S}^{\,i+}(x) \ge \max\left(LS^{i+}_f(y), LS^{i-}_f(y)\right) \cdot e^{-\beta \cdot d(y,x)}\end{aligned}$$

holds, and $\forall x \in D^n : \tilde{S}^{\,i+}(x) \ge \tilde{S}^{*\,i+}_{f,\beta}(x)$ is derived accordingly. In a similar manner, we can show that $\forall x \in D^n$: $\tilde{S}^{\,i-}(x) \ge \tilde{S}^{*\,i-}_{f,\beta}(x)$ holds. ☐

Theorem 2. *For all $i \in [m]$, $\hat{S}^{*\,i+}_{f,\beta}$ and $\hat{S}^{*\,i-}_{f,\beta}$ are enhanced direction-oriented β-smooth upper bound on $\hat{LS}^{i+}_f$ and $\hat{LS}^{i-}_f$. In addition, for all $x \in D^n$, $\hat{S}^{*\,i\pm}_{f,\beta}(x) \le \hat{S}^{\,i\pm}(x)$ holds for every enhanced direction-oriented β-smooth upper bound $\hat{S}^{\,i+}$ and $\hat{S}^{\,i-}$ on $\hat{LS}^{i+}_f$ and $\hat{LS}^{i-}_f$.*

Proof. The proof is very similar to that of Theorem 1. ☐

Theorem 3. *For all $x \in D^n$ and $i \in [m]$, $\hat{S}^{*\,i+}_{f,\beta}(x) \le \tilde{S}^{*\,i+}_{f,\beta}(x) \le S^{*}_{f,\beta}(x) \ \wedge\ \hat{S}^{*\,i-}_{f,\beta}(x) \le \tilde{S}^{*\,i-}_{f,\beta}(x) \le S^{*}_{f,\beta}(x)$ holds.*

Proof. Note that $\forall x \in D^n : \hat{LS}^{i+}_f(x) \le LS^{i+}_f(x) \le LS_f(x) \ \wedge\ \hat{LS}^{i-}_f(x) \le LS^{i-}_f(x) \le LS_f(x)$ holds from the definition and discussion in Section IV.A. Therefore,

$$
\begin{aligned}
\hat{S}^{*\,i+}_{f,\beta}(x) &= \max\Bigg(\hat{LS}_f^{i+}(x),\ \hat{LS}_f^{i-}(x)\cdot e^{-2\beta},\\
&\max_{y\in D^n, y\neq x}\left(\max\left(\hat{LS}_f^{i+}(y),\ \hat{LS}_f^{i-}(y)\right)\cdot e^{-\beta\cdot d(y,x)}\right)\Bigg)\\
&\le \max\Bigg(LS_f^{i+}(x),\ LS_f^{i-}(x)\cdot e^{-2\beta},\\
&\max_{y\in D^n, y\neq x}\left(\max\left(LS_f^{i+}(y),\ LS_f^{i-}(y)\right)\cdot e^{-\beta\cdot d(y,x)}\right)\Bigg)\\
&\left(= \quad \tilde{S}^{*\,i+}_{f,\beta}(x)\right)\\
&\le \max\left(LS_f(x), \max_{y\in D^n, y\neq x}\left(LS_f(y)\cdot e^{-\beta\cdot d(y,x)}\right)\right)\\
&= \max_{y\in D^n}\left(LS_f(y)\cdot e^{-\beta\cdot d(y,x)}\right) = S^*_{f,\beta}(x).
\end{aligned}
$$

$\hat{S}^{*\,i-}_{f,\beta}(x) \le \tilde{S}^{*\,i-}_{f,\beta}(x) \le S^*_{f,\beta}(x)$ holds true in a similar manner. □

Lemma 3. *(ϵ-differentially private mechanism using an enhanced direction-oriented smooth upper bound based on improved DOLS)*
Let h be a symmetric (α,β)-admissible noise probability density function on $\mathbb{R}$, where α and β are linear functions with respect to the input and $\alpha(0)=\beta(0)=0$. Define h^+ and h^- as follows:

$$
h^+(z) = \begin{cases} 2\cdot h(z) & (z\ge 0)\\ 0 & (z<0)\end{cases}
$$
$$
\textit{and} \quad h^-(z) = \begin{cases} 0 & (z\ge 0)\\ 2\cdot h(z) & (z<0)\end{cases} .
$$

Let z_i^+ and z_i^- be random variables derived from h^+ and h^-, respectively, for each $i\in[m]$ independently. Set $\alpha=\alpha(\epsilon)$ and $\beta=\beta\left(\frac{\epsilon}{m}\right)$. For a function $f: D^n\to\mathbb{R}^m$, let $\hat{S}^{\,i+}, \hat{S}^{\,i-}: D^n\to\mathbb{R}$ be enhanced direction-oriented β-smooth upper bound on $\hat{LS}_f^{i+}$ and $\hat{LS}_f^{i-}$ for each $i\in[m]$. Thereafter, the mechanism M such that

$$
\begin{aligned}
&\forall i:\ M(x)_i\\
&= \quad f(x)_i + \begin{cases} \frac{\hat{S}^{\,i+}(x)}{\alpha}\cdot z_i^+ & \text{(with a probability of 1/2)}\\ \frac{\hat{S}^{\,i-}(x)}{\alpha}\cdot z_i^- & \text{(with a probability of 1/2)}\end{cases}
\end{aligned}
$$

is ϵ-differentially private.

Proof. It is sufficient to show that for all x,y s.t. $d(x,y)=1$,

$$
\forall T\in\mathbb{R}^m: \Pr[M(x)=T] \le e^{\epsilon}\cdot\Pr[M(y)=T].
$$

Here, $\Pr[M(x)=T]=\prod_{i\in[m]}\Pr[M(x)_i=T_i]$, and

$$
\Pr[M(x)_i=T_i] = \begin{cases} \frac{1}{2}\cdot\Pr\left[z_i^+ = \frac{T_i-f(x)_i}{N^{\,i+}(x)}\right] & (T_i\ge f(x)_i)\\ \frac{1}{2}\cdot\Pr\left[z_i^- = \frac{T_i-f(x)_i}{N^{\,i-}(x)}\right] & (T_i< f(x)_i)\end{cases},
$$

where $N^{\,i\pm}(x) := \frac{\hat{S}^{\,i\pm}(x)}{\alpha}$.

(I) When $f(x)_i\le f(y)_i$:
(i) When $T_i<f(x)_i$:

$$
\begin{aligned}
&\Pr[M(x)_i=T_i]\\
&= \frac{1}{2}\cdot\Pr\left[z_i^- = \frac{T_i-f(x)_i}{N^{\,i-}(x)}\right]\\
&\le \frac{1}{2}\cdot e^{\frac{\epsilon}{2m}}\cdot e^{\frac{f(y)_i-f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}\cdot\epsilon}\cdot\Pr\left[z_i^- = \frac{T_i-f(y)_i}{N^{\,i-}(y)}\right]\\
&\Bigg[\because \frac{N^{\,i-}(x)}{N^{\,i-}(y)}\le e^{\beta\left(\frac{\epsilon}{m}\right)},\\
&\quad \frac{f(x)_i-f(y)_i}{N^{\,i-}(y)} = -\alpha\cdot\left(\frac{f(y)_i-f(x)_i}{\hat{S}^{\,i-}(y)}\cdot\epsilon\right)\Bigg]\\
&= e^{\frac{\epsilon}{2m}}\cdot e^{\frac{f(y)_i-f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}\cdot\epsilon}\cdot\Pr[M(y)_i=T_i].
\end{aligned}
$$

(ii) When $f(x)_i\le T_i<f(y)_i$:

$$
\begin{aligned}
&\Pr[M(x)_i=T_i]\\
&= \frac{1}{2}\cdot\Pr\left[z_i^+ = \frac{T_i-f(x)_i}{N^{\,i+}(x)}\right]\\
&\le \frac{1}{2}\cdot e^{\frac{\epsilon}{2m}}\cdot e^{\frac{f(y)_i-f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}\cdot\epsilon}\cdot\Pr\left[z_i^+ = \frac{f(y)_i-T_i}{N^{\,i-}(y)}\right]\\
&\Bigg[\because \frac{N^{\,i+}(x)}{N^{\,i-}(y)}\le e^{\beta\left(\frac{\epsilon}{m}\right)},\\
&\quad \frac{f(x)_i+f(y)_i-2T_i}{N^{\,i-}(y)} \le \alpha\left(\frac{f(y)_i-f(x)_i}{\hat{S}^{\,i-}(y)}\cdot\epsilon\right),\\
&\quad \frac{f(x)_i+f(y)_i-2T_i}{N^{\,i-}(y)} > -\alpha\left(\frac{f(y)_i-f(x)_i}{\hat{S}^{\,i-}(y)}\cdot\epsilon\right)\Bigg]\\
&= \frac{1}{2}\cdot e^{\frac{\epsilon}{2m}}\cdot e^{\frac{f(y)_i-f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}\cdot\epsilon}\cdot\Pr\left[z_i^- = \frac{T_i-f(y)_i}{N^{\,i-}(y)}\right]\\
&= e^{\frac{\epsilon}{2m}}\cdot e^{\frac{f(y)_i-f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}\cdot\epsilon}\cdot\Pr[M(y)_i=T_i].
\end{aligned}
$$

(iii) When $f(y)_i\le T_i$:

$$
\begin{aligned}
\Pr[M(x)_i=T_i] &= \frac{1}{2}\cdot\Pr\left[z_i^+ = \frac{T_i-f(x)_i}{N^{\,i+}(x)}\right]\\
&\le \frac{1}{2}\cdot\Pr\left[z_i^+ = \frac{T_i-f(y)_i}{N^{\,i+}(x)}\right]\\
&[\because 0\le T_i-f(y)_i<T_i-f(x)_i]\\
&\le \frac{1}{2}\cdot e^{\frac{\epsilon}{2m}}\cdot\Pr\left[z_i^+ = \frac{T_i-f(y)_i}{N^{\,i+}(y)}\right]\\
&\left[\because \frac{N^{\,i+}(x)}{N^{\,i+}(y)}\le e^{\beta\left(\frac{\epsilon}{m}\right)}\right]\\
&= e^{\frac{\epsilon}{2m}}\cdot\Pr[M(y)_i=T_i].
\end{aligned}
$$

Therefore, the following relation holds:

$$
\Pr[M(x)_i=T_i]\le e^{\frac{\epsilon}{2m}}\cdot e^{\frac{f(y)_i-f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}\cdot\epsilon}\cdot\Pr[M(y)_i=T_i].
$$

(II) When $f(y)_i<f(x)_i$:
In a similar manner to (I), the following relation holds:

$$
\Pr[M(x)_i=T_i]\le e^{\frac{\epsilon}{2m}}\cdot e^{\frac{f(x)_i-f(y)_i}{2\cdot\hat{S}^{\,i+}(y)}\cdot\epsilon}\cdot\Pr[M(y)_i=T_i].
$$

From (I) and (II),

$$\begin{aligned}&\Pr[M(x)_i = T_i]\\ &\le e^{\frac{\epsilon}{2m}} \cdot e^{\max\left(\frac{f(y)_i - f(x)_i}{2\cdot \hat{S}^{i-}(y)}, \frac{f(x)_i - f(y)_i}{2\cdot \hat{S}^{i+}(y)}\right)\cdot\epsilon} \cdot \Pr[M(y)_i = T_i]\end{aligned}$$

holds. Here, from the characteristic of the improved DOLS,

$$\begin{aligned}&\sum_{i\in[m]} \max\left(\frac{f(y)_i - f(x)_i}{2\cdot \hat{S}^{i-}(y)}, \frac{f(x)_i - f(y)_i}{2\cdot \hat{S}^{i+}(y)}\right)\\ &\le \sum_{i\in[m]} \max\left(\frac{\max_{j\in[n]} c_j^{i-}(y)}{2\cdot \hat{LS}_f^{i-}(y)}, \frac{\max_{j\in[n]} c_j^{i+}(y)}{2\cdot \hat{LS}_f^{i+}(y)}\right) \le \frac{1}{2}.\end{aligned}$$

Consequently, we can show that

$$\begin{aligned}\Pr[M(x) = T] &= \prod_{i\in[m]} \Pr[M(x)_i = T_i]\\ &\le e^{\frac{\epsilon}{2}} \cdot e^{\frac{\epsilon}{2}} \cdot \prod_{i\in[m]} \Pr[M(y)_i = T_i]\\ &= e^{\epsilon}\cdot \Pr[M(y) = T].\end{aligned}$$

□

Theorem 4. *For all $\beta > 0$ and $i \in [m]$,*

$$\begin{aligned}\hat{S}^{*\,i\pm}_{f,\beta}(x) \;=\; \max\Bigg(&\hat{LS}_f^{i\pm}(x),\ \hat{LS}_f^{i\mp}(x)\cdot e^{-2\beta},\\ &\max_{y\in D^n, 1\le d(y,x)\le \frac{1}{\beta}\cdot\ln\left(\frac{GS_f}{\max\left(\hat{LS}_f^{i+}(x), \hat{LS}_f^{i-}(x)\right)}\right)+2}\\ &\left(\max\left(\hat{LS}_f^{i+}(y), \hat{LS}_f^{i-}(y)\right)\cdot e^{-\beta\cdot d(y,x)}\right)\Bigg). \quad (1)\end{aligned}$$

Proof. When $d(y,x) > \frac{1}{\beta}\cdot\ln\left(\frac{GS_f}{\max\left(\hat{LS}_f^{i+}(x), \hat{LS}_f^{i-}(x)\right)}\right) + 2$,

$$e^{-\beta\cdot d(y,x)} < \frac{\max\left(\hat{LS}_f^{i+}(x), \hat{LS}_f^{i-}(x)\right)}{GS_f}\cdot e^{-2\beta}.$$

From this and that $\max\left(\hat{LS}_f^{i+}(y), \hat{LS}_f^{i-}(y)\right) \le GS_f$,

$$\begin{aligned}&\max\left(\hat{LS}_f^{i+}(y), \hat{LS}_f^{i-}(y)\right)\cdot e^{-\beta\cdot d(y,x)}\\ <\ &\max\left(\hat{LS}_f^{i+}(x), \hat{LS}_f^{i-}(x)\right)\cdot e^{-2\beta}\\ \le\ &\max\left(\hat{LS}_f^{i\pm}(x), \hat{LS}_f^{i\mp}(x)\cdot e^{-2\beta}\right).\end{aligned}$$

Consequently, from Definition 13, the relation (1) holds. □

Theorem 5. *We denote $K(x) := \max\left(\hat{LS}_f^{i+}(x), \hat{LS}_f^{i-}(x)\right)$. For all β satisfying*

$$\forall i\in[m]:\ e^{\beta} \ge \max_{x,x':d(x,x')=1} \frac{K(x')}{K(x)}, \quad (2)$$

the following equality holds for all $x \in D^n$:

$$\max_{y:y\neq x}\left(K(y)\cdot e^{-\beta\cdot d(y,x)}\right) = \max_{y:d(y,x)=1}\left(K(y)\cdot e^{-\beta}\right). \quad (3)$$

Furthermore, when

$$e^{\beta} \ge \max\left(\frac{\hat{LS}_f^{i\mp}(x)}{\hat{LS}_f^{i\pm}(x)},\ 1\right)\cdot \max_{x,x':d(x,x')=1}\frac{K(x')}{K(x)}, \quad (4)$$

$\hat{S}^{*\,i\pm}_{f,\beta}(x) = \hat{LS}_f^{i\pm}(x)$.

Proof. We first show that when (2) holds,

$$\max_{y,d(y,x)=1}\left(K(y)\cdot e^{-\beta}\right) \ge K(y')\cdot e^{-\beta\cdot d(y',x)}$$

for all y' s.t. $d(y',x) > 1$.

(I) When $d(y',x) = 2$, there exists some y'' satisfying $d(y',y'') = 1 \ \wedge\ d(y'',x) = 1$. Therefore,

$$\begin{aligned}\max_{y,d(y,x)=1}\left(K(y)\cdot e^{-\beta}\right) &\ge K(y'')\cdot e^{-\beta}\\ &\ge \left(K(y')\cdot e^{-\beta}\right)\cdot e^{-\beta} \quad [\because (2)]\\ &= K(y')\cdot e^{-\beta\cdot d(y',x)}.\end{aligned}$$

(II) We assume that $\max_{y,d(y,x)=1}\left(K(y)\cdot e^{-\beta}\right) \ge K(y'')\cdot e^{-\beta\cdot d(y'',x)}$ holds for all y'' satisfying $d(y'',x) = k$. Here, for any y' s.t. $d(y',x)$, there exists some y'' satisfying $d(y',y'') = 1 \ \wedge\ d(y'',x) = k$. Therefore, as in case (I),

$$\max_{y,d(y,x)=1}\left(K(y)\cdot e^{-\beta}\right) \ge K(y')\cdot e^{-\beta\cdot d(y',x)}.$$

From (I) and (II), the relation (3) holds for all x.

Furthermore, when (4) holds,

$$\begin{aligned}\max_{y,d(y,x)=1}\left(K(y)\cdot e^{-\beta}\right) &\le K(x)\cdot \frac{1}{\max\left(\frac{\hat{LS}_f^{i\mp}(x)}{\hat{LS}_f^{i\pm}(x)},\ 1\right)}\\ &= \hat{LS}_f^{i\pm}(x).\end{aligned}$$

In addition, $\hat{LS}_f^{i\pm}(x) \ge \hat{LS}_f^{i\mp}(x)\cdot e^{-2\beta}$ holds. Consequently, $\hat{S}^{*\,i\pm}_{f,\beta}(x) = \hat{LS}_f^{i\pm}(x)$. □

Lemma 4. (*Unbiased ϵ-differentially private mechanism using an enhanced direction-oriented smooth upper bound based on improved DOLS*)

Let h be a symmetric (α,β)-admissible noise probability density function on $\mathbb{R}$, where α and β are linear functions with respect to the input and $\alpha(0) = \beta(0) = 0$. Define h^+ and h^- as follows:

$$h^+(z) = \begin{cases} 2\cdot h(z) & (z\ge 0)\\ 0 & (z<0)\end{cases}$$

$$\textit{and}\quad h^-(z) = \begin{cases} 0 & (z\ge 0)\\ 2\cdot h(z) & (z<0)\end{cases}.$$

Let z_i^+ and z_i^- be random variables derived from h^+ and h^-, respectively, for each $i\in[m]$ independently. Set $\alpha = \alpha(\epsilon)$ and $\beta = \beta\left(\frac{\epsilon}{m}\right)$. For a function $f : D^n \to \mathbb{R}^m$, let $\hat{S}^{i+}, \hat{S}^{i-} : D^n \to \mathbb{R}$ be enhanced direction-oriented β-smooth upper bound on $\hat{LS}_f^{i+}$ and $\hat{LS}_f^{i-}$ for each $i \in [m]$. Thereafter, the mechanism M such that

$$\forall i: \ M(x)_i = f(x)_i + \begin{cases} \frac{\hat{S}^{\,i+}(x)}{\alpha} \cdot z_i^+ & \left(\text{with a probability of } \frac{\hat{S}^{\,i-}(x)}{\hat{S}^{\,i+}(x)+\hat{S}^{\,i-}(x)}\right) \\ \frac{\hat{S}^{\,i-}(x)}{\alpha} \cdot z_i^- & \left(\text{with a probability of } \frac{\hat{S}^{\,i+}(x)}{\hat{S}^{\,i+}(x)+\hat{S}^{\,i-}(x)}\right) \end{cases}$$

is $(\epsilon + 2m\beta)$*-differentially private.*

Proof. It is sufficient to show that for all x, y s.t. $d(x, y) = 1$,

$$\forall T \in \mathbb{R}^m: \ \Pr[M(x) = T] \le e^{\epsilon+2m\beta} \cdot \Pr[M(y) = T].$$

Here, $\Pr[M(x) = T] = \prod_{i\in[m]} \Pr[M(x)_i = T_i]$, and

$$\Pr[M(x)_i = T_i] = \begin{cases} \frac{\hat{S}^{\,i-}(x)}{\hat{S}^{\,i+}(x)+\hat{S}^{\,i-}(x)} \cdot \Pr\left[z_i^+ = \frac{T_i - f(x)_i}{N^{\,i+}(x)}\right] & (T_i \ge f(x)_i) \\ \frac{\hat{S}^{\,i+}(x)}{\hat{S}^{\,i+}(x)+\hat{S}^{\,i-}(x)} \cdot \Pr\left[z_i^- = \frac{T_i - f(x)_i}{N^{\,i-}(x)}\right] & (T_i < f(x)_i) \end{cases},$$

where $N^{\,i\pm}(x) := \frac{\hat{S}^{\,i\pm}(x)}{\alpha}$.

(I) When $f(x)_i \le f(y)_i$:

(i) When $T_i < f(x)_i$:

$$\begin{aligned} &\frac{\hat{S}^{\,i+}(x) + \hat{S}^{\,i-}(x)}{\hat{S}^{\,i+}(x)} \cdot \Pr[M(x)_i = T_i] \\ &= \Pr\left[z_i^- = \frac{T_i - f(x)_i}{N^{\,i-}(x)}\right] \\ &\le e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}\cdot\epsilon} \cdot \Pr\left[z_i^- = \frac{T_i - f(y)_i}{N^{\,i-}(y)}\right] \\ &= \frac{\hat{S}^{\,i+}(y) + \hat{S}^{\,i-}(y)}{\hat{S}^{\,i+}(y)} \cdot e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}\cdot\epsilon} \cdot \Pr[M(y)_i = T_i]. \end{aligned}$$

(ii) When $f(x)_i \le T_i < f(y)_i$:

$$\begin{aligned} &\frac{\hat{S}^{\,i+}(x) + \hat{S}^{\,i-}(x)}{\hat{S}^{\,i-}(x)} \cdot \Pr[M(x)_i = T_i] \\ &= \Pr\left[z_i^+ = \frac{T_i - f(x)_i}{N^{\,i+}(x)}\right] \\ &\le e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}\cdot\epsilon} \cdot \Pr\left[z_i^+ = \frac{f(y)_i - T_i}{N^{\,i-}(y)}\right] \\ &= e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}\cdot\epsilon} \cdot \Pr\left[z_i^- = \frac{T_i - f(y)_i}{N^{\,i-}(y)}\right] \\ &= \frac{\hat{S}^{\,i+}(y) + \hat{S}^{\,i-}(y)}{\hat{S}^{\,i+}(y)} \cdot e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}\cdot\epsilon} \cdot \Pr[M(y)_i = T_i]. \end{aligned}$$

(iii) When $f(y)_i \le T_i$:

$$\begin{aligned} &\frac{\hat{S}^{\,i+}(x) + \hat{S}^{\,i-}(x)}{\hat{S}^{\,i-}(x)} \cdot \Pr[M(x)_i = T_i] \\ &= \Pr\left[z_i^+ = \frac{T_i - f(x)_i}{N^{\,i+}(x)}\right] \\ &\le \Pr\left[z_i^+ = \frac{T_i - f(y)_i}{N^{\,i+}(x)}\right] \\ &\le e^{\frac{\epsilon}{2m}} \cdot \Pr\left[z_i^+ = \frac{T_i - f(y)_i}{N^{\,i+}(y)}\right] \\ &= \frac{\hat{S}^{\,i+}(y) + \hat{S}^{\,i-}(y)}{\hat{S}^{\,i-}(y)} \cdot e^{\frac{\epsilon}{2m}} \cdot \Pr[M(y)_i = T_i]. \end{aligned}$$

Here, the following inequalities hold:

$$\frac{\hat{S}^{\,i+}(y) + \hat{S}^{\,i-}(y)}{\hat{S}^{\,i+}(x) + \hat{S}^{\,i-}(x)} \le e^{\beta},$$
$$\frac{\hat{S}^{\,i+}(x)}{\hat{S}^{\,i+}(y)} \le e^{\beta}, \ \frac{\hat{S}^{\,i-}(x)}{\hat{S}^{\,i+}(y)} \le e^{\beta}, \ \frac{\hat{S}^{\,i-}(x)}{\hat{S}^{\,i-}(y)} \le e^{\beta}.$$

Therefore, the following relation holds:

$$\begin{aligned} &\Pr[M(x)_i = T_i] \\ &\le e^{2\beta} \cdot e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}\cdot\epsilon} \cdot \Pr[M(y)_i = T_i]. \end{aligned}$$

(II) When $f(y)_i < f(x)_i$:

In a similar manner to (I), the following relation holds:

$$\begin{aligned} &\Pr[M(x)_i = T_i] \\ &\le e^{2\beta} \cdot e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(x)_i - f(y)_i}{2\cdot\hat{S}^{\,i+}(y)}\cdot\epsilon} \cdot \Pr[M(y)_i = T_i]. \end{aligned}$$

From (I) and (II),

$$\begin{aligned} &\Pr[M(x)_i = T_i] \\ &\le e^{\frac{\epsilon}{2m}} \cdot e^{\max\left(\frac{f(y)_i - f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}, \frac{f(x)_i - f(y)_i}{2\cdot\hat{S}^{\,i+}(y)}\right)\cdot\epsilon} \cdot \Pr[M(y)_i = T_i] \end{aligned}$$

holds. Here, from the characteristic of the improved DOLS,

$$\begin{aligned} &\sum_{i\in[m]} \max\left(\frac{f(y)_i - f(x)_i}{2\cdot\hat{S}^{\,i-}(y)}, \frac{f(x)_i - f(y)_i}{2\cdot\hat{S}^{\,i+}(y)}\right) \\ &\le \sum_{i\in[m]} \max\left(\frac{\max_{j\in[n]} c_j^{\,i-}(y)}{2\cdot\hat{LS}_f^{\,i-}(y)}, \frac{\max_{j\in[n]} c_j^{\,i+}(y)}{2\cdot\hat{LS}_f^{\,i+}(y)}\right) \le \frac{1}{2}. \end{aligned}$$

Consequently, we can show that

$$\begin{aligned} \Pr[M(x) = T] &= \prod_{i\in[m]} \Pr[M(x)_i = T_i] \\ &\le e^{2m\beta} \cdot e^{\frac{\epsilon}{2}} \cdot e^{\frac{\epsilon}{2}} \cdot \prod_{i\in[m]} \Pr[M(y)_i = T_i] \\ &= e^{\epsilon+2m\beta} \cdot \Pr[M(y) = T]. \end{aligned}$$

□

Theorem 6. *Let M be the mechanism described in Lemma 4, and the following relation holds:*

$$\forall x \in D^n : \forall i : \mathbb{E}[M(x)_i] = f(x)_i.$$

Proof. For any $x \in D^n$ and i,

$$\begin{aligned} \mathbb{E}[M(x)_i] &= f(x)_i + \frac{\hat{S}^{\,i-}(x)}{\hat{S}^{\,i+}(x) + \hat{S}^{\,i-}(x)} \cdot \frac{\hat{S}^{\,i+}(x)}{\alpha} \cdot \mathbb{E}[z_i^+] \\ &\quad + \frac{\hat{S}^{\,i+}(x)}{\hat{S}^{\,i+}(x) + \hat{S}^{\,i-}(x)} \cdot \frac{\hat{S}^{\,i-}(x)}{\alpha} \cdot \mathbb{E}[z_i^-] \\ &= f(x)_i. \quad [\because \mathbb{E}[z_i^+] = -\mathbb{E}[z_i^-]] \end{aligned}$$

□